

Kynurenine-Centered Metabolic-Immune Trap: the Dual-Arm Model

TCGA discovery (n = 123) → METABRIC validation (n = 320) → single-cell & spatial resolution → dual prediction frameworks

UPSTREAM INFLAMMATORY ARM (IFN- γ -induced)

- IFN- γ drives IDO1 / TDO2 transcriptional activation
- CD8+ T-cell-rich, inflamed microenvironment
- IDO1 high → favorable OS (HR 0.78, p = 0.0018)
- Checkpoint-blockade-responsive (TIDE / IMPRES)

TCGA C3 (n = 50) · METABRIC
M2+M3 (n = 223)

KYNURERENINE- CENTERED TRAP

Score-expression discordance
Kyn score $\uparrow$ · IDO1 $\downarrow$
CD8 / cytotoxicity $\downarrow$

DOWNSTREAM METABOLIC ARM (IFN- γ -independent)

- KP program weight $\uparrow$ (relative ssGSEA ranking)
- Upstream enzymes low; downstream AFMID / AADAT / GPT2 up-trending
- Myeloid KP program: IL411 / KYNU / KMO in macrophages & DCs (T cells < 1.5%)
- TAM M2-linked exclusion (TIDE, program-level)

TCGA C1 (n = 28) · METABRIC
M1 (n = 97) · OS HR 1.49

VALIDATION CHAIN

Consensus clustering (k = 3) → dual-track sensitivity → swap-in (0%) / uniform-metric (100%) clustering
→ expression-matched null (p = 0.002) → multivariable arbitration (subtype p = 8.7×10^{-17}) → scRNA (n = 100,064)
→ spatial Visium (n = 11,877 spots) → METABRIC replication → TIDE ± IMPRES

CANDIDATE IHC STRATIFICATION PANEL (research-stage)

CD8 + IDO1 = operative core: IDO1 high → inflammation-induced type (ICB benefit) | both low → KP-program-defined trap
KP downstream nodes (AFMID / AADAT / GPT2) = exploratory, protein-level validation pending